

1. Answer all questions in a neat and precise manner.
2. Only calculators are allowed, no other material is allowed.
3. All parts of the question must be answered at one place.
4. Make appropriate assumptions as required.

Answer the following questions:

1. The spectra of two message signals $m_1(t)$ and $m_2(t)$ are occupying same bandwidth, having RMS value of their amplitudes as 5 V and 10 V respectively. If these two signals are to be modulated over a single carrier signal with peak amplitude of 10 V with carrier frequency of 10 MHz.

(i) Determine what type of amplitude modulation scheme is more suitable and explain the process of modulation/demodulation using a neat block diagram, along with suitable math. (7 Marks)

(ii) Also determine the power contained in the modulated signal across that the load resistor $200\ \Omega$, assuming that f_{m1} and f_{m2} are frequencies of $m_1(t)$ and $m_2(t)$ respectively, and $f_{m1} > f_{m2}$. (3 Marks)

2. A complex message signal $m(t) = 6 \cos 2\pi 1000t + 12 \cos 2\pi 2500t + 8 \cos 2\pi 5000t$ volts is modulating a carrier signal $c(t) = 8 \cos 2\pi 10^5 t$ volts.

(i) Determine the AM signal (DSB-FC). (2 Marks)

(ii) Determine the modulation index of the complex AM signal. (2 Marks)

(iii) Was the AM signal distorted? if yes, what is the minimum required measure that has to be taken to take care of distortion. Determine the new value of the modulation index after removing the distortion. (2 Marks)

(iv) Draw the voltage vs frequency spectrum of the AM signal after removing the distortion. (2 Marks)

(v) Assuming a load resistance of $100\ \Omega$, determine the power of the AM signal before and after removing the distortion. (2 Marks)

3. A two-stage amplifier circuit has following parameters: Input signal power: $-30\ \text{dBm}$ & Input noise power: $-60\ \text{dBm}$, Power gain of Stage 1 and Stage 2: $20\ \text{dB}$ each, and the Noise figure of Stage 1: $1.25\ \text{dB}$ & Noise figure of Stage 2: $2.25\ \text{dB}$.

Calculate the following:

(i) SNR at the input of amplifier stage 1. (2 Marks)

(ii) SNR at the input of amplifier stage 2. (2 Marks)

(iii) SNR at the output of amplifier stage 2. (2 Marks)

(iv) Overall reduction in input SNR to output SNR. (2 Marks)

(v) Overall noise figure of the amplifier circuit. (2 Marks)